Anna Oliveras Tous

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PROFILE

PhD researcher in generative AI for medical imaging at the University of Barcelona and Eurecat, supervised by Prof. Petia Radeva and Dr. Vicent Ribas. I develop diffusion models that synthesise 3D medical images (CT and MRI) and forecast how a patient's scan will evolve over time, with work published in Scientific Reports and accepted at WACV 2027. Background in signal processing, machine learning and deep learning from an MSc in Computer Vision, a BSc in Telecommunications Engineering, industry experience in medical computer vision, and research internships in Barcelona and Lausanne.

PROFESSIONAL EXPERIENCE

AI Researcher in Digital Health **Dec 2024 – Present**
Eurecat, Digital Health Unit *Barcelona, Spain*

- Develop generative models (pixel-space and latent diffusion) that synthesise medical images, mainly 3D chest CT, conditioned on anatomy and clinical metadata to create new training data and forecast tumour growth.
- Built LAND, an anatomically guided latent diffusion model for high-resolution 3D chest CT (Scientific Reports, 2026).
- Built MUMINS, which predicts a patient's follow-up scan from a baseline scan with a per-voxel uncertainty map, validated on lung CT and brain MRI (WACV 2027).
- Trained 3D models both on low-memory GPUs (resource-constrained) and multi-GPU on HPC clusters.

Computer Vision Engineer **Nov 2023 – Nov 2024**
Intek Medical *Barcelona, Spain*

- Developed AI software for automated hair and skin analysis, supporting personalised, patient-specific treatments.
- Applied deep learning to 3D human reconstruction to obtain anthropometric measurements.
- Designed AI-assisted data acquisition and annotation tools, used to build a large dataset for model training.

Research Intern **May 2023 – Sep 2023**
Computer Vision Center (CVC) *Barcelona, Spain*

- Applied Large Language Models for Document Visual Question Answering using PyTorch and Hugging Face.
- Improved performance by adding spatial layout information (bounding boxes) to the prompt.
- Added the visual modality through a Vision Transformer so the language model could interpret visual tokens.

Research Intern **Jun 2022 – Aug 2022**
LTS5, EPFL *Lausanne, Switzerland*

- Worked on the "3T-study" (Test, Triage and Treat) cervical cancer screening programme led by the Geneva University Hospitals (HUG) and the Dschang District Hospital, Cameroon.
- Developed deep learning models to detect strong motion in smartphone videos recorded during VIA screening.

EDUCATION

PhD in Computer Science **Dec 2024 – Present**
University of Barcelona (UB) and Eurecat *Barcelona, Spain*

- Thesis: generative models for synthetic medical imaging, mainly CT and MRI.
- Supervisors: Prof. Petia Radeva (UB, AIBA group) and Dr. Vicent Ribas (Eurecat).

Master's degree in Computer Vision **Sep 2022 – Sep 2023**
Autonomous University of Barcelona (UAB) *Barcelona, Spain*

- Relevant courses: Machine Learning, Deep Learning, 3D Vision, Video analysis
- Grade: 8.9/10. Top 5 MCV List Award (top 5 GPA of the programme)

Bachelor's degree in Telecommunications Engineering **Sep 2018 – Jun 2022**
Polytechnic University of Catalonia (UPC) *Barcelona, Spain*

- Relevant courses: Signal processing, Audiovisual processing, Networks, Programming
- Grade: 7.9/10. Top 10% of the class
- Thesis at LTS5, EPFL (Prof. Jean-Philippe Thiran): image segmentation for cervical cancer screening, presented at MICAD 2022. Grade: 5/5.

PUBLICATIONS

MUMINS: Metadata-conditioned Uncertainty-aware Medical Image Next-state Synthesis **2027**
A. Oliveras, R. Marí, R. Redondo, O. Guardià-Olivella, C. I. Ugwu, A. Tost, B. Nagarajan, C. Migliorelli, V. Ribas, P. Radeva
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), accepted | arXiv:2609.17169 | Code

Anatomically Guided Latent Diffusion for High-Resolution 3D Chest CT Synthesis **2026**
A. Oliveras, R. Marí, R. Redondo, O. Guardià, C. I. Ugwu, A. Tost, B. Nagarajan, C. Migliorelli, V. Ribas, P. Radeva
Scientific Reports (Nature Portfolio), 2026 | doi:10.1038/s41598-026-51634-4 | Project page

TECHNICAL SKILLS

Machine learning: Diffusion models, generative models, uncertainty estimation, foundation model training and fine-tuning, Vision Transformers, LLMs and vision-language models

Training at scale: PyTorch DDP, FSDP, mixed precision, gradient checkpointing, PyTorch Lightning

Medical imaging: 3D CT and MRI, DICOM, NIfTI, MONAI, SimpleITK, nibabel, 3D Slicer

HPC and MLOps: Slurm, Docker, Singularity, Linux, Weights & Biases, TensorBoard, MLflow

Programming: Python, C/C++, Java, MATLAB, Shell scripting, HTML5/CSS

Libraries and tools: PyTorch, Hugging Face, TensorFlow, Keras, OpenCV, Scikit-Learn, NumPy, Pandas, PyQt, Git/GitHub, Blender, MeshLab

Data and cloud: SQL, MongoDB, Microsoft Azure

LANGUAGES

Catalan (native), Spanish (native), English (C2, Cambridge C2 Proficiency), French (A2)

DISTINCTIONS

- Top 5 MCV List Award: *top 5 GPA in the Master's in Computer Vision (UAB)* Sep 2023
- Recognition for the university entrance exams (PAU) Jun 2018

HOBBIES

Viola (professional conservatory studies), cycling, trail running and skiing